COMMERCIAL ITEM DESCRIPTION

TESTER, ANTIFREEZE AND BATTERY-ELECTROLYTE SOLUTION

The General Services Administration has authorized the use of this commercial item description as a replacement for Military Specification MIL-T-28679 which will be canceled.

This commercial item description covers a portable antifreeze and battery-electrolyte tester, designed for rapid and accurate testing of ethylene glycol antifreeze solutions and battery-electrolyte solutions.

Salient characteristics.

The tester shall be a lightweight, hand-held, monocular, optical instrument, designed to measure refractive index. It shall be calibrated to provide scales for direct reading of specific gravity of a lead-acid battery-electrolyte solution, and the freezing point of an ethylene glycol antifreeze solution, on a graduated transparent reticle through a telescopic, recessed eyepiece. Shall be a transmission type designed to provide maximum contrast between light and dark. Shall be designed to automatically compensate for temperature variations.

The tester shall consist of a tubular shaped body containing an eyepiece, a reticle, a measuring prism and a hinged cover plate. A clear plastic syringe for withdrawing an antifreeze sample, and a plastic dipstick for obtaining an electrolyte sample, shall be provided with and stored on or in the tester body.

The body shall be constructed of die cast aluminum covered with a high impact plastic housing. The housing shall have a suitable cutout at the distal end of the instrument to facilitate storage by being suspended from a hook.

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The telescopic, recessed eyepiece at the proximal end of the tester, shall magnify the scales on the reticle for ease of readability. The eyepiece shall be constructed in a manner to allow reading of the reticle without interference while wearing corrective eyeglasses.

The reticle shall be of transparent glass and etched to show, as a minimum, the following scales and markings:

The battery-electrolyte condition scale of the reticle shall show graduations in degrees of change (specific gravity) of a lead-acid battery as:

1.250 through 1.300	"GOOD"
1.200 through 1.250	"FAIR"
1.100 through 1.200	"RECHARGE"

The antifreeze protection scale of the reticle shall show the freezing point of ethylene glycol solutions in both degrees Fahrenheit (°F) and degrees Centigrade (°C). The degree marks on the Fahrenheit scale shall be graduated in 5°F intervals from -50°F to +32°F. The degree marks for the Centigrade scale shall be graduated in 3°C intervals from -45°C to 0°C.

The measuring prism shall be of optically clear glass bonded to the tester body to form a measuring window at the distal end of the instrument. The window shall be the specimen sample holder, and the entrance for the beam of light entrained through the optical system. The bonding material shall be resistant to the electrolyte and ethylene glycol test solutions.

The hinged cover plate shall be fabricated of clear plastic that shall permit admittance of light to the optical system when closed over the measuring window. An opening at the top of the cover shall allow placement of sample specimen so that the liquid is drawn between the cover and the measuring window surface during test.

Shall be capable of testing in daylight or in other illumination such as automobile headlights.

When any special or important procedure is to be followed in operating or servicing the equipment, the unit shall be equipped with an instruction plate, suitably located, describing such procedure.

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A copy of a service manual shall be provided with the tester.

Calibration verification. The tester shall be considered accurately calibrated when reticle readings correspond with known freezing point and specific gravity readings. The sample unit shall be tested as follows:

A reading of +32° ± 2°F (0° ± 1°C) shall show on the reticle when distilled water at +32°F (0°C) is tested at the measuring window.

A reading of -30° ± 2°F (-34.5 ± 1°C) shall show on the reticle scale when a calibrated sucrose solution of 31.8 percent by weight (refractive index 1.3884 at +68°F (+20°C) is used at +68°F (+20°C).

Workmanship. The tester shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One tester, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties)of the item.

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Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for inspections and tests. Sampling for inspections and tests shall be conducted in accordance with commercial practice. The unit of product for test purposes shall be one tester.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its contractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

6630-00-105-1418

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:
DoD - MB

Agent:
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53426	TESTER, ANTIFREEZE AND BATTERY-ELECTROLYTE SOLUTION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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